# BudgetMesh: Rate-Aware Training and Compression for Opaque Differentiable Meshes

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## Abstract

*In this work, we study how differentiable opaque meshes, as introduced by MeshSplatting, can be compressed. They render in any engine, but a trained scene occupies hundreds of megabytes. Unlike Gaussian splats, they cannot be simplified after training without a large loss in quality: because every pixel is covered by a single opaque triangle, removing vertices from a trained mesh degrades quality far more than training a smaller mesh from the start. We argue that the right place to decide the size of a mesh is during its growth, while the optimizer can still adapt. Building on this idea, we introduce BudgetMesh, a complete compression scheme spanning training, encoding and decoding: a rate-aware growth rule that charges each refinement step its true cost in encoded bytes, a closed-loop controller that steers growth to a user-specified file size, and a dedicated codec with a standalone decoder that writes one compact, engine-ready file. Built on the MeshSplatting pipeline, BudgetMesh produces a more efficient mesh than its native densification at the same vertex count. Given a target file size, the output file lands within about 10% of it on four datasets. At 20 to 5 MB the delivered files are 31–122× smaller than the trained model at an average cost of 0.3–1.3 dB.*

## 1. Introduction

Differentiable rendering can now optimize an *opaque triangle mesh* directly from posed images [18, 20]: a set of triangles with per-vertex position, color and view-dependent appearance, drawn by an ordinary rasterizer. Unlike Gaussian splats [12], such a mesh renders in any engine and on any phone, with depth and collisions for free. Its weakness is size. A trained MeshSplatting [18] scene occupies 430–780 MB in 32-bit floats, while compressed Gaussian scenes reach 5–20 MB [3, 15]. No compression method exists for this representation, and the obvious ones fail.

The first obvious route is to train the mesh and then simplify it, as one does with any mesh. For opaque meshes this does not work: every pixel is covered by exactly one triangle, so each vertex is placed and colored for the neighbours it has, and a decimator leaves the survivors where they were. At the same vertex count, a mesh trained to that size from the start is better than a trained-then-simplified mesh on 62 of 63 scene–budget pairs, by 1.85 dBon average (Sec. 4.1). The second route is a standard mesh codec. Draco [7] and MPEG TFAN [17] assume clean manifold meshes with a few attributes per vertex; a MeshSplatting mesh is a stitched soup with 45 appearance coefficients per vertex, and both codecs produce files 3–18× larger than ours (Sec. 4.4). The size of the file therefore has to be decided *while the mesh is trained*. Today it cannot be: MeshSplatting's vertex count emerges from its densification heuristics, a vertex cap fixes vertices rather than bytes, and the same training setting yields files that differ by 2.5× across scenes, so reaching a given size means training each scene several times.

We present BudgetMesh, which takes a target file size in megabytes and returns a file of that size after one training run. The idea is to make bytes part of every decision that creates or removes a primitive. BudgetMesh splits a triangle only if the rendering error it carries is worth the bytes the split adds to the encoded file, and collapses an edge only if the error around it is worth less than the bytes the collapse frees; one exchange rate, the price of a byte, governs both. The price is not a hyperparameter. From the target size, BudgetMesh computes how many vertices the file can hold, with byte costs measured by running the actual encoder on the mesh during training, plans the growth toward that count, and sets the price at each growth step so that exactly the planned number of splits is admitted; after growth, the same rule run in reverse trims the mesh until the measured size meets the target. A codec designed for this representation then writes one file, and a standalone decoder turns it into a standard indexed mesh.

On Mip-NeRF 360, Tanks and Temples, Deep Blending and NeRF-Synthetic, 83 of 89 files land within ±10% of the requested size, at a quality cost below 0.1 dBwhere it can be measured directly against a sweep of fixed prices. At 20, 10 and 5 MB the files are 31×, 60× and 122× smaller than the trained model at −0.30, −0.66 and −1.32 dB. Our key contributions are:

- We propose BudgetMesh, the first method that takes the file size of a differentiable opaque mesh as an input. A single byte price governs both growth and collapse dur-

ing training, and the price is derived from the target size through byte costs measured by the encoder itself.
- We introduce a codec for this representation, with implicit vertex order, edge-cache connectivity coding, neighbour-predicted positions and a two-level appearance dictionary with neighbour-context indices; it is 2.7–3.5× smaller than Draco carrying the same payload.
- We show that opaque meshes must be sized during training, and evaluate the full system on four datasets against simplification after training, MeshSplatting with a vertex cap, generic mesh codecs and compressed Gaussian representations.

## 2. Related Work

**Meshes from posed images.** MobileNeRF [4], BakedSDF [27] and Binary Opacity Grids [21] extract a mesh from a trained neural field and bake appearance into textures, at 100–900 MB per scene. Meshes extracted from Gaussian scenes [8, 28, 29] reach 1 GB and more. MeshSplatting [18] and MiLo [20] optimize the mesh itself. None of these controls the file size; MeshSplatting's only small variant is an RGB-only file that drops view dependence.

**Compressing Gaussian scenes.** Pruning, vector quantization and learned entropy models reduce 3DGS [12] from 700 MB to 5–50 MB [2, 3, 5, 14–16, 25]. Two properties make this easier than our setting: Gaussians are independent primitives, so a trained scene can be pruned cheaply, and the rate is set by a hyperparameter swept per scene. Opaque meshes have neither property.

**Mesh compression and simplification.** Edgebreaker [22], shipped in Draco [7], and MPEG TFAN [17] code manifold connectivity at about 2 bits per triangle and predict attributes along the traversal. Edge collapse with quadric error [6, 10] is the standard way to reduce a mesh. Our split and collapse rule follows the Lagrangian rate–distortion principle of video coding [23]; what is new is where it acts, inside the optimization that creates the primitives, with the rate measured by the final encoder and the multiplier derived from a target size.

## 3. Method

BudgetMesh keeps MeshSplatting's renderer, losses and schedule and changes how the mesh grows and shrinks so that the encoded file meets a target size $T$. Sec. 3.1 fixes notation, Sec. 3.2 gives the rule that decides every split and collapse, Sec. 3.3 explains how the price in that rule is obtained from $T$ (Alg. 1), and Sec. 3.4 describes the codec whose bytes the rule counts.

### 3.1. Preliminaries

MeshSplatting [18] represents a scene by $V$ vertices, each with a position $\mathbf{x} \in \mathbb{R}^3$, a DC color $\mathbf{c} \in \mathbb{R}^3$ and 45 spherical-harmonic (SH) coefficients $\mathbf{h}$ of degrees 1–3, and $F$ triangles over them. Training runs 30k iterations. Every 500 iterations until 15k it *densifies*: it samples triangles by an importance score and replaces each by four, inserting the three edge midpoints as new vertices; it also prunes triangles that render with low opacity. At 11k a restricted Delaunay step stitches the soup into a connected mesh, an opacity floor then rises until every triangle is opaque, and a final cleaning removes triangles that cover no pixel. We keep all of this. Let $R$ be the size of the encoded file in bytes and $D$ the rendering error on the training views. We write $D_t$ for the *error of triangle* $t$: the per-pixel $L_1$ residual summed over the pixels whose front-most triangle is $t$ (the rasterizer returns this id), averaged over the views rendered since the last decision and normalized per megapixel.

### 3.2. One price for splits and collapses

We want the mesh that minimizes $D + \lambda R$ for a price $\lambda$ (error units per byte). A local change is worth making when it lowers this sum, which gives two rules with the same $\lambda$.

**Split.** Splitting $t$ adds three vertices and three faces, so it adds $\Delta R_{\text{split}} = 3(b_v + b_f)$ bytes, where $b_v$ and $b_f$ are the file's bytes per vertex and per face. We use $D_t$ as the error the split can remove and split $t$ iff

$$D_t \,/\, \Delta R_{\text{split}} > \lambda. \tag{1}$$

Since $\Delta R_{\text{split}}$ is the same for every triangle at one step, Eq. (1) ranks triangles by error and $\lambda$ sets how many are split. This replaces MeshSplatting's importance sampling: a triangle is refined because it is wrong, not because it is visible.

**Collapse.** Collapsing edge $(u, v)$ merges $u$ and $v$ into one vertex and deletes the faces on the edge. It frees $\Delta R_{uv} = b_v\,[\,u, v \text{ survive cleaning}\,] + b_f\, n_{uv}$ bytes, with $n_{uv}$ the deleted faces that would survive cleaning, and it risks the error $D_{uv}$ of the faces around $u$ and $v$. We collapse the edges with the lowest $D_{uv}/\Delta R_{uv}$, all of them below $\lambda$, as an independent set (no two collapses share a vertex) with a fold-over check. The merged vertex is placed at the midpoint and takes the average of the two attribute vectors and of their optimizer moments, so training continues without a restart. This is what simplification after training cannot offer: the survivors keep being optimized.

### 3.3. From a target size to the price

A fixed $\lambda$ does not determine a file size: across our scenes one price lands at files 2.5× apart. BudgetMesh therefore sets $\lambda$ at every step from $T$ (Alg. 1).

**Byte probe.** The file size is affine in the mesh, $R = b_v V + b_f F + c$, where $c$ covers the dictionaries and the header;

$b_v$ and $b_f$ vary between scenes and operating points (8–13 bytes per vertex), so they are measured, not assumed. Every 1000 iterations once MeshSplatting starts pruning (4k), a copy of the mesh is cleaned exactly as the final mesh will be and passed through the real encoder of Sec. 3.4. The probe returns $R$, the coefficients $b_v$, $b_f$, $c$, the faces per vertex $f$, and the fraction of vertices that survive cleaning; these replace the previous values.

**Growth plan.** The vertex count the target buys is

$$V^\star = (\alpha T - c) \,/\, (b_v + f\, b_f). \tag{2}$$

With $m$ growth steps left, the plan raises $V$ geometrically, $V \to V\,(V^\star/V)^{1/m}$, which asks for $k = V\big((V^\star/V)^{1/m} - 1\big)/(3\sigma)$ splits now: each split adds three vertices, of which a fraction $\sigma$ survives MeshSplatting's pruning at the next step ($\sigma$ is measured from the previous cohort). The plan is recomputed at every step, so errors in one step are spread over the steps that remain.

**Price.** $\lambda$ is set to the $k$-th largest ratio $D_t/\Delta R_{\text{split}}$, so that Eq. (1) admits $k$ triangles. Because the errors are re-measured between steps, the number that actually passes differs from $k$; we divide $k$ by the measured yield $y$ (passed over asked) of the last steps before taking the rank.

**Trimming.** From 12.5k to 19.5k, every 500 iterations, the probe measures $R$ and a least-squares fit of the probe history extrapolates it to the end of the opacity ramp. If the estimate exceeds $T$, edges are collapsed by the rule of Sec. 3.2 until the excess is removed, at most 15% of the vertices per round, and $\lambda$ becomes the score of the last edge collapsed. If it does not, nothing is collapsed, so trimming never pushes the file below the target.

**Growth aim.** After the mesh is stitched at 11k, the set of visible faces keeps growing as the opacity floor rises, by 5–40% depending on the capture class, and none of it can be observed before 11k. The plan therefore aims above the target by a factor $\alpha$ (1.25 for captured scenes, 1.1 for synthetic objects) and leaves the correction to trimming, which can only remove bytes. $\alpha$ is the one calibrated constant of BudgetMesh.

**Algorithm 1** BudgetMesh training for a target size $T$

**Require:** images, target $T$, aim $\alpha$
initialize MeshSplatting; byte table $(b_v, b_f, f, c)$ from defaults
**for** iteration $i = 1 \dots 30$k **do**
  MeshSplatting optimization step; accumulate $D_t$ ▷ per-triangle error
  **if** growth step ($i \le 15$k) **then**
    probe every 1k: encode a cleaned copy → update table
    $V^\star \leftarrow (\alpha T - c)/(b_v + f b_f)$ ▷ Eq. 2
    $k \leftarrow$ geometric plan toward $V^\star$, survival $\sigma$
    $\lambda \leftarrow (k/y)$-th largest $D_t/\Delta R_{\text{split}}$
    split all $t$ with $D_t/\Delta R_{\text{split}} > \lambda$ ▷ Eq. 1
  **if** trimming step (12.5k $\le i \le$ 19.5k) **then**
    $\hat{R} \leftarrow$ probe, extrapolated to end of ramp
    **if** $\hat{R} > T$ **then**
      collapse lowest $D_{uv}/\Delta R_{uv}$ until $\hat{R} \le T$
      $\lambda \leftarrow$ score of last collapsed edge
final cleaning; encode (Sec. 3.4) → one file

## 3.4. Codec

The trained mesh stores 52 floats per vertex and three 32-bit indices per face. The codec makes three lossy reductions, then codes the result losslessly in an order where each stream is predicted from what the decoder already has.

**Lossy reductions.** (i) Positions are quantized on a per-axis uniform grid of $b = \lceil \log_2(\text{extent}/(0.1\,\bar{e})) \rceil$ bits, $\bar{e}$ the median edge length (14–18 bits on our scenes): the step is a tenth of a typical edge; one bit finer costs bytes for no PSNR, one bit coarser loses 0.06–0.2 dB. (ii) The 45 SH coefficients are 87% of the floats, and neighbouring vertices on a surface share their view-dependent behaviour, but the coefficients are too noisy to predict from one another. We therefore replace them by a two-level per-scene dictionary, $\mathbf{h}_v \approx \mathbf{B}[i_v] + \mathbf{B}_r[j_v]$, with $|\mathbf{B}| = 4096$ and $|\mathbf{B}_r| = 256$ entries fitted by $k$-means and refined for 400 steps against the rendering loss with the assignments fixed. Each vertex keeps two indices. (iii) The DC color is quantized to 7 bits per channel, and the opacity, uniform after training, is one scalar.

**Vertex order.** Faces are reordered into a cache-friendly strip order [11], and vertex ids are assigned in order of first reference. The decoder can then infer every id: a vertex that has not appeared yet is always the next id. No permutation is transmitted, and every later stream can use the already-decoded neighbours of a vertex.

**Connectivity.** The coder keeps the 64 most recently decoded directed edges. A face that shares an edge with one of them is coded as *hit*: the cache slot, the edge direction and the third vertex; otherwise as *miss*: three vertices. A vertex is coded as 0 when it is new and as the difference to the last coded id otherwise. All symbols are coded with an adaptive binary arithmetic coder whose contexts are the symbol type and the bits already coded; no tables are sent. Edgebreaker-style coders traverse a manifold; our meshes are stitched soups with 45% boundary edges and thousands of components, which forces Edgebreaker to split vertices, so a local cache is the better model here. Connectivity costs about 15 bits per face and is the largest stream (39–43% of the file).

**Positions.** Each quantized position is predicted by its earliest already-decoded mesh neighbour, rebuilt by the decoder from the faces, and the residual is arithmetic coded per axis with the magnitude of the previous residual as context: about 16 bits per vertex against 45 raw.

**Appearance indices.** Index frequencies are nearly uniform, so an order-0 entropy coder gains nothing; the redundancy

Table 1. **Deciding the size during training.** PSNR gain in dB, mean over pairs, with the number of pairs where the first method is better. (a) MeshSplatting trained with a vertex cap vs. the same FP32 model simplified to that vertex count after training. (b) BudgetMesh vs. MeshSplatting trained with a vertex cap set from BudgetMesh's vertex count: at the same vertex count before any entropy coding and after our codec, and at the same file size. Three counts / sizes per scene.

(a) trained to the count vs. simplified after training

| | pairs | gain | range |
|---|---|---|---|
| Mip-360 + T&T | 33 | +2.07 (33) | +0.10 … +5.48 |
| Deep Blending | 6 | +2.71 (6) | +1.39 … +4.72 |
| NeRF-Synthetic | 24 | +1.33 (23) | −0.03 … +4.35 |

(b) BudgetMesh vs. MeshSplatting with a vertex cap

| | same $V$, FP32 | same $V$, coded | same size |
|---|---|---|---|
| Mip-360 + T&T | +0.27 (29/30) | +0.26 (29/30) | +0.21 (30/33) |
| NeRF-Synthetic | +0.24 (15/15) | +0.22 (15/15) | +0.12 (16/24) |

Deep Blending in (b): same file size only, +0.08 (4/6); see text.

is spatial. For vertex $v$ the coder lists the distinct indices of its already-decoded neighbours and codes one flag (is $i_v$ among them) and, if so, its position in the list; otherwise the index itself. The DC colors are byte-shuffled and compressed with zstd.

**Decoding.** The decoder reads the connectivity, rebuilds the neighbour graph from it, then decodes positions and indices and expands $\mathbf{h}_v$ from the dictionaries. It needs nothing but the file, runs in 2–6 s on one CPU core, and outputs a standard indexed mesh that an ordinary vertex shader draws. The encoder verifies a byte-exact round trip before writing; all loss comes from the three reductions above.

# 4. Experiments

**Datasets and protocol.** Mip-NeRF 360 [1] (9 scenes), Tanks and Temples [13] (truck, train), NeRF-Synthetic [19] (8 objects) and Deep Blending [9] (drjohnson, playroom). Deep Blending was not used while developing the method; every number on it comes from one run with the settings fixed on the other datasets. MeshSplatting training is unchanged (30k iterations, images downsampled by 2 indoors and by 4 outdoors, the standard test split, black background for NeRF-Synthetic). Quality is PSNR, SSIM and LPIPS of the *decoded file* rendered at the renderer's supersampling; all sizes are bytes of the single delivered file. The *FP32 model* is the trained MeshSplatting mesh in 32-bit floats, our reproduction; it is the reference for compression ratios and quality loss. One configuration for all scenes: appearance dictionary $4096 + 256$, position grid of 14–18 bits by scene extent, 7-bit DC color, growth aim $\alpha = 1.25$ for captured scenes and 1.1 for synthetic objects, collapse rounds every 500 iterations from 12.5k to 19.5k. "At the same vertex count" means after MeshSplatting's final cleaning.

## 4.1. Size must be decided during training

Table 1 answers the two questions a reader will ask first. *Why not decimate a trained mesh?* For each scene and three vertex counts (those of BudgetMesh's own 20 / 10 / 5 MB files; 4 / 2.5 / 1.5 MB on NeRF-Synthetic) we compare MeshSplatting trained with that count as a vertex cap against the FP32 model simplified to the same count by edge collapse (quadric error with an appearance term) followed by the same 3000-step fine-tuning, both through our codec. Training to the count wins on 62 of 63 pairs, by 1.85 dBon average and by up to 5.5 dB; the gap grows as the budget shrinks (0.5 dBat the largest, 3.8 dBat the smallest count on real scenes). The simplified meshes keep fewer faces per vertex and show holes and speckle (supplement). *Is the growth rule itself better than a vertex cap?* Against MeshSplatting trained with the cap set from BudgetMesh's own vertex count (pairs within $\pm 15\%$), rate-aware growth is better on 44 of 45 pairs, by 0.27 dBon real scenes and 0.24 dBon objects, measured on the FP32 meshes *before any entropy coding*; the margin is unchanged after the codec, so it belongs to the growth rule, not to the encoder. At the same file size after the codec the margin is 0.21 dBon real scenes (30/33); on NeRF-Synthetic it is 0.12 dB(16/24) because the 4 MB group contains three objects at their capacity limit (Sec. 4.2); at 1.5 MB it is 0.24 dBon 8 of 8. On Deep Blending the vertex-cap runs exist only at other vertex counts, so we report the same-file-size comparison alone: +0.08 dB, 4 of 6.

## 4.2. Size as an input

Table 2 reports, per dataset and target, the mean file size, the size error $(\text{file size} - \text{target})/\text{target}$ and the count of scenes within $\pm 10\%$; Fig. 1 draws the same runs as rate–distortion curves, with the per-scene spread of the landing as whiskers at each target. Across the four datasets 83 of 89 runs land within $\pm 10\%$ with one setting per capture class. The misses are systematic and we report them: indoor scenes land 5–14% above the target (kitchen at 10 MB is the worst, $+13.5\%$), because the visible-face set keeps growing after the growth phase and the collapse rounds can remove only so much per round; outdoor scenes at 20 MB land up to 7% below, because the last growth step admits fewer splits than planned and nothing can add vertices afterwards. On NeRF-Synthetic several objects cannot fill 3–6 MB at all: the price falls to its floor from the first steps and the file stays small (ficus, mic; six of eight objects at 6 MB). These *capacity-limited* runs are counted separately and never hidden. Deep Blending, untouched during development, lands at +7 to $+10\%$ on average: the same indoor behaviour as Mip-NeRF 360, no retuning.

**Cost of control.** The best a growth rule can do at a given file size is what it reaches when the size is *not* prescribed: the same rule run at a fixed price, several trainings per

Table 2. **Size control and quality per dataset and target.** Mean file size; size error $(\text{file size} - T)/T$ as the mean and the range over scenes; scenes within $\pm 10\%$; PSNR of the decoded file and its change against the FP32 model; SSIM and LPIPS. Means over the scenes below their capacity limit; capacity-limited objects (the file cannot grow to $T$) are counted in parentheses. The 15, 7.5 and 3 MB targets are in the supplement.

| dataset | target $T$ | scenes | file size | size error | range | within $\pm 10\%$ | PSNR ($\Delta$) | SSIM | LPIPS |
|---|---|---|---|---|---|---|---|---|---|
| Mip-NeRF 360 + T&T | 20 MB | 11 | 19.9 MB | $-0.5\%$ | $-6.6 \ldots +10.0\%$ | 10 | 23.75 ($-0.30$) | 0.712 | 0.397 |
| | 10 MB | 11 | 10.2 MB | $+2.0\%$ | $-4.7 \ldots +13.5\%$ | 10 | 23.39 ($-0.66$) | 0.684 | 0.442 |
| | 5 MB | 11 | 5.0 MB | $+0.6\%$ | $-4.8 \ldots +9.2\%$ | 11 | 22.72 ($-1.32$) | 0.644 | 0.491 |
| Deep Blending | 20 MB | 2 | 21.3 MB | $+6.7\%$ | $+3.6 \ldots +9.7\%$ | 2 | 26.94 ($-0.28$) | 0.838 | 0.452 |
| | 10 MB | 2 | 11.0 MB | $+10.2\%$ | $+8.3 \ldots +12.0\%$ | 1 | 26.54 ($-0.69$) | 0.830 | 0.478 |
| | 5 MB | 2 | 5.4 MB | $+7.9\%$ | $+4.7 \ldots +11.1\%$ | 1 | 25.89 ($-1.34$) | 0.817 | 0.502 |
| NeRF-Synthetic | 4 MB | 6 (+2) | 3.9 MB | $-2.9\%$ | $-10.3 \ldots +3.4\%$ | 5 | 30.60 ($-0.13$) | 0.942 | 0.082 |
| | 2.5 MB | 7 (+1) | 2.6 MB | $+2.6\%$ | $-6.6 \ldots +9.2\%$ | 7 | 30.81 ($-0.30$) | 0.947 | 0.076 |
| | 1.5 MB | 8 | 1.6 MB | $+4.0\%$ | $+1.0 \ldots +9.0\%$ | 8 | 30.36 ($-0.88$) | 0.944 | 0.080 |

scene, the size read off afterwards. Against that reference curve (three fixed prices per scene, landing at about 5, 12 and 25 MB on real scenes and 1.5, 2.5 and 4 MB on objects), BudgetMesh is 0.04 dBlower at 20 MB, 0.06 dBat 2.5 MB and 0.02 dBat 1.5 MB, where a fixed-price run lands at about the same size as the target, and 0.13 dBlower at 10 MB, where the curve is read between two prices (the curve is concave, so this chord reading under-estimates the cost; the three points bound the error at 0.19 dBon average over a span twice as wide). The cost of getting the size in one training is therefore below 0.1 dBwhere it can be measured directly and below about 0.3 dBeverywhere. Run-to-run variance of the full method, three seeds on four scenes: the file size moves by 0.5–5% of the target and the PSNR by 0.01–0.14 dB. The reference also shows why a price is not a size: at one fixed price the file size varies by 2.2–2.5× across the 11 real scenes (4.7–11.7 MB at one price, 10.8–24.0 MB at another).

### 4.3. Compression and other representations

Against the FP32 model the 20, 10 and 5 MB files are 31×, 60× and 122× smaller at $-0.30$, $-0.66$ and $-1.32$ dBon real scenes (Table 2); against the same meshes written as a quantized PLY and compressed with zstd they are 7×,

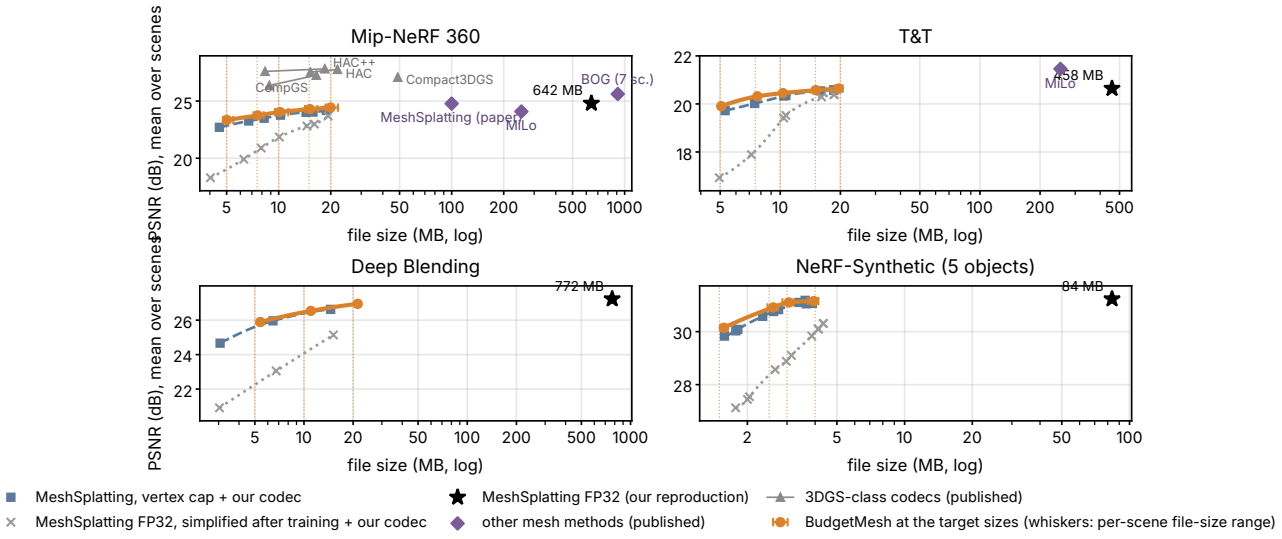



Figure 1. **Rate–distortion per dataset.** Mean PSNR vs. mean file size. Orange: BudgetMesh; dotted lines mark the targets, whiskers the per-scene range of the file size. Blue: MeshSplatting with a vertex cap, same codec. Grey crosses: the FP32 model simplified after training, same codec. Star: FP32 model. Diamonds and triangles: published mesh and 3DGS-class methods at their own sizes. NeRF-Synthetic: the five objects below their capacity limit.

Table 3. **Against Gaussian codecs and mesh-output methods**, dataset means; size in MB. Gaussian rows are the published HAC++ benchmark [3]. Mesh rows: (a) published numbers collected in [18], where MiLo, RaDe-GS and GOF report one asset size, for Mip-NeRF 360 (n/r: not reported); (b) MobileNeRF and NeRF2Mesh as published; MeshSplatting reports no NeRF-Synthetic results, so its FP32 rows are our reproduction. BudgetMesh: $T$ = 20 / 10 / 5 MB on scenes, 4 / 2.5 / 1.5 MB on objects. ‡Black background; published NeRF-Synthetic rows use white. Deep Blending is in the supplement.

**(a)** Mip-NeRF 360 and Tanks and Temples

| | Mip-NeRF 360 | | | | Tanks and Temples | | | |
|---|---|---|---|---|---|---|---|---|
| method | PSNR | SSIM | LPIPS | MB | PSNR | SSIM | LPIPS | MB |
| 3DGS [12] | 27.46 | 0.812 | 0.222 | 750.9 | 23.69 | 0.844 | 0.178 | 431.0 |
| Scaffold-GS [16] | 27.50 | 0.806 | 0.252 | 253.9 | 23.96 | 0.853 | 0.177 | 86.50 |
| Compact3DGS [14] | 27.08 | 0.798 | 0.247 | 48.80 | 23.32 | 0.831 | 0.201 | 39.43 |
| LightGaussian [5] | 27.00 | 0.799 | 0.249 | 44.54 | 22.83 | 0.822 | 0.242 | 22.43 |
| HAC [2], low | 27.53 | 0.807 | 0.238 | 15.26 | 24.04 | 0.846 | 0.187 | 8.10 |
| HAC, high | 27.77 | 0.811 | 0.230 | 21.87 | 24.40 | 0.853 | 0.177 | 11.24 |
| ContextGS [26], low | 27.62 | 0.808 | 0.237 | 12.68 | 24.20 | 0.852 | 0.184 | 7.05 |
| ContextGS, high | 27.75 | 0.811 | 0.231 | 18.41 | 24.29 | 0.855 | 0.176 | 11.80 |
| HAC++ [3], low | 27.60 | 0.803 | 0.253 | 8.34 | 24.22 | 0.849 | 0.190 | 5.18 |
| HAC++, high | 27.82 | 0.811 | 0.231 | 18.48 | 24.32 | 0.854 | 0.178 | 8.63 |
| GOF mesh [28] | 20.78 | 0.573 | 0.465 | 1500 | 21.69 | 0.690 | 0.326 | n/r |
| RaDe-GS mesh [29] | 23.56 | 0.668 | 0.361 | 1100 | 20.51 | 0.659 | 0.344 | n/r |
| MiLo [20] | 24.09 | 0.688 | 0.323 | 253 | 21.46 | 0.706 | 0.348 | n/r |
| MeshSplatting FP32 [18] | 24.81 | 0.731 | 0.309 | 642 | 20.64 | 0.760 | 0.272 | 459 |
| BudgetMesh, $T$=20 MB | 24.44 | 0.704 | 0.411 | 20.0 | 20.64 | 0.748 | 0.337 | 19.6 |
| BudgetMesh, $T$=10 MB | 24.04 | 0.673 | 0.458 | 10.2 | 20.45 | 0.731 | 0.368 | 10.3 |
| BudgetMesh, $T$=5 MB | 23.35 | 0.632 | 0.508 | 5.0 | 19.91 | 0.696 | 0.416 | 5.1 |

**(b)** NeRF-Synthetic‡

| | 8 objects | | | |
|---|---|---|---|---|
| method | PSNR | SSIM | LPIPS | MB |
| 3DGS [12] | 33.80 | 0.970 | 0.031 | 68.46 |
| Scaffold-GS [16] | 33.41 | 0.966 | 0.035 | 19.36 |
| Compact3DGS [14] | 33.33 | 0.968 | 0.034 | 5.54 |
| LightGaussian [5] | 32.73 | 0.965 | 0.037 | 7.84 |
| HAC [2], low | 33.24 | 0.967 | 0.037 | 1.18 |
| HAC, high | 33.71 | 0.968 | 0.034 | 1.86 |
| ContextGS [26], low | 32.79 | 0.965 | 0.040 | 1.01 |
| ContextGS, high | 33.51 | 0.968 | 0.035 | 1.56 |
| HAC++ [3], low | 33.03 | 0.966 | 0.039 | 0.88 |
| HAC++, high | 33.76 | 0.969 | 0.033 | 1.84 |
| MobileNeRF [4] | 30.90 | 0.947 | 0.062 | 125.8 |
| NeRF2Mesh [24] | 29.76 | 0.940 | 0.072 | 73.5 |
| MeshSplatting FP32‡ | 31.24 | [TBD: ] | [TBD: ] | 65.7 |
| BudgetMesh ‡, $T$=4 MB | 31.13 | 0.951 | 0.070 | 3.49 |
| BudgetMesh ‡, $T$=2.5 MB | 30.97 | 0.950 | 0.072 | 2.49 |
| BudgetMesh ‡, $T$=1.5 MB | 30.36 | 0.944 | 0.080 | 1.56 |

14× and 27× smaller. The ratio splits into two factors: the file holds fewer vertices (1.4×, 3.0× and 6.7× fewer than the 1.9–3.2M of the FP32 model), the growth rule's share, and each vertex costs 11–13 bytes instead of 231, the codec's share. Within the mesh class (Table 3a) BudgetMesh is the only entry below 250 MB, 0.4 dBbelow its own FP32 model at 20 MB and above MiLo, RaDe-GS and GOF at 13–75× smaller files; Binary Opacity Grids reports 25.62 dBat 913 MB on a 7-scene subset, without SSIM or LPIPS. Against compressed Gaussians (Table 3a), at 5–20 MB HAC++ and HAC are 3.3–4 dBhigher in PSNR on Mip-NeRF 360 and Tanks and Temples (Deep Blending in the supplement), 0.10–0.11 higher in SSIM and 0.16–0.21 lower in LPIPS; on NeRF-Synthetic (Table 3b) the gap is 2.5–3 dB, with the caveat that our objects are rendered on black and the published ones on white. Within the mesh class on NeRF-Synthetic, BudgetMesh at 2.5 MB matches MobileNeRF's PSNR at 1/50 of its size and is 1.2 dBabove NeRF2Mesh at 1/30 of its size. That gap is the representation's: opaque triangles against anisotropic Gaussians. What the mesh buys is a standard indexed mesh that renders at 145–350 fps on an iPhone 17 Pro at capture resolution and decodes in 2.3–6.4 s on one CPU core. LPIPS degrades faster than PSNR as $T$ shrinks; Sec. 4.5 traces this to the appearance dictionary.

Table 4. **Generic codecs on the same mesh.** File size of each route divided by ours, mean and range over the 11 Mip-NeRF 360 / T&T scenes, and the largest PSNR loss against FP32 (dB). The three "our payload" rows carry exactly our quantized positions, indices and DC, so their quality equals ours.

| route | size ÷ ours | range | loss |
|---|---|---|---|
| Draco, 8-bit SH | 15.0× | 12.3–17.7 | 0.21 |
| quantized PLY + zstd | 5.2× | 4.4–6.2 | 0.21 |
| MPEG TFAN, 8-bit SH* | 3.7× | 3.0–4.2 | 0.21 |
| Draco, our payload | 3.0× | 2.7–3.5 | ours |
| Draco geometry + zstd payload | 1.4× | 1.3–1.6 | ours |
| Draco geometry alone | 1.05× | 0.85–1.24 | – |
| Draco, RGB only | 2.1× | 1.8–2.4 | 4.4 |
| baked texture + Draco | 2.7× | 2.4–3.2 | 4.4 |
| ours | 1× | – | 0.8 |

* 10 scenes. RGB-only routes drop view dependence.

### 4.4. Against generic mesh codecs

Could an off-the-shelf mesh codec replace ours? Table 4 runs the same FP32 meshes through Draco [7], the MPEG TFAN coder [17] and zstd, and compares bytes with our file for the same mesh (our encoder with all vertices kept, −0.1 to −0.8 dB). Keeping view dependence as 8-bit SH attributes, Draco needs 12–18× and MPEG 3–4× our bytes; the only generic routes near our size drop the SH altogether (RGB-only vertex colors, or a baked texture) and lose 0.7–

Table 5. **Ablation.** Size error (mean) and scenes within $\pm 10\%$; PSNR at the same file size relative to the full method, read on each scene's reference curve of three fixed prices, and the scenes where the variant is better. Ficus is capacity-limited at 2.5 MB and excluded.

| variant | size error | $\pm 10\%$ | $\Delta$PSNR | better |
|---|---|---|---|---|
| *Mip-NeRF 360 + T&T, $T = 10$ MB, 11 scenes* | | | | |
| full BudgetMesh | $+2.0\%$ | 10 | 0 | – |
| no collapse | $+4.5\%$ | 9 | $-0.03$ | 4 |
| no byte probe | $+5.0\%$ | 10 | $+0.03$ | 9 |
| MeshSplatting split selection | $-27\%$ | 0 | $-0.42$ | 0 |
| *NeRF-Synthetic, $T = 2.5$ MB, 7 objects* | | | | |
| full BudgetMesh | $+2.6\%$ | 7 | 0 | – |
| no collapse | $+9.0\%$ | 2 | $-0.05$ | 3 |
| no byte probe | $+12.8\%$ | 1 | $+0.07$ | 6 |
| MeshSplatting split selection | $-11\%$ | 4 | $-0.24$ | 1 |

4.4 dB. The generic rows are near-lossless and ours is not, so the table measures the bytes each coder needs for essentially the same mesh rather than a rate–distortion curve; The fair generic point is Draco carrying exactly our appearance payload, the two dictionary indices and the 7-bit DC color as one generic attribute on the same positions and faces: its file is 2.7–3.5$\times$ ours on every real-scene asset (2.1–3.5$\times$ on objects), and Draco's geometry alone, positions and connectivity with no appearance at all, already weighs 0.85–1.24$\times$ our whole file. Our appearance streams cost 18–42% of our file; coded by Draco's attribute predictor they cost two to three files. Our meshes are triangle soups stitched late in training and are not manifold, which is what the Edgebreaker-style coders pay for; the mesh statistics are in the supplement.

### 4.5. Where the bytes go

Splitting a file into its streams (figure in the supplement): connectivity is 39–43% and lossless, positions 20%, appearance indices 17–20%, DC color 11–13%, and the dictionaries a fixed 0.65 MB (3% of a 20 MB file, 12% of a 5 MB one). Two findings follow. *Appearance sharing carries the loss:* 45 of the 52 floats per vertex are SH, replaced by two indices; at the top operating point an 8-bit instead of 7-bit DC color recovers only 0.04–0.08 dBof the 0.4–0.6 dBloss on indoor scenes, so the remainder is the dictionary, and so is the LPIPS increase of Table 3. *Dictionary size should follow vertex count:* over a sweep of dictionary sizes and prices on all 21 scenes, the fixed $4096 + 256$ is within 0.16 dBof the best choice at every point on captured scenes, but on the two smallest objects (mic, ficus) a 1024-entry dictionary at a lower price is $+0.5$–0.6 dBat the same file size. The sweep is in the supplement; the paper uses the fixed setting.

### 4.6. Ablation

Table 5 switches off one part of the training rule at a time on all 11 real scenes at 10 MB and all NeRF-Synthetic objects at 2.5 MB, same code, codec and evaluation (ficus is capacity-limited at 2.5 MB and excluded). At 10–20 MB on real scenes the growth plan alone lands within 10% on 9–10 of 11 scenes and none of the parts changes the quality at the same file size by more than 0.03 dB; the collapse rounds and the byte probe each remove one failure mode. The collapse trims the indoor overshoot (kitchen $+20\% \rightarrow +13\%$; on objects it takes the hit rate from 2/7 to 7/7). The probe is what makes small files land: with the fixed constants the 1.5 MB objects land $+18\%$ on average, 1 of 8 within 10%, against $+4\%$ and 8 of 8 with the probe, while at 20 MB the constants are within 10% on 10 of 11. The last row is the one that supports Sec. 4.1: with the number of splits per step copied from the full run, choosing the triangles by MeshSplatting's importance score instead of error per byte gives meshes that MeshSplatting's own pruning then cuts further, and at the same file size they are 0.42 dBworse on every real scene and 0.24 dBworse on objects.

### 4.7. Qualitative results

The supplement shows decoded files at the three main targets next to the ground truth and the FP32 model, with an enlarged crop and the error map per column, and the same vertex count reached three ways on one view; the simplified mesh shows holes and speckle where the trained one does not.

**Limitations.** The growth aim is one calibrated constant per capture class; it stands in for the growth of the visible-face set after stitching, which cannot be observed earlier. Indoor scenes land 5–14% above the target and outdoor scenes at 20 MB up to 7% below. Small objects cannot fill large targets. Trimming (Sec. 3.3) is one-sided: it protects against undershoot from collapsing, not against the growth plan admitting too few splits. Opaque meshes are 3–4 dBbelow Gaussian codecs at the same file size. The entropy coders are fixed; a learned rate model is future work.

## 5. Conclusion

Opaque differentiable meshes cannot be shrunk after training, so BudgetMesh decides their size while they grow: a growth rule priced in encoded bytes, a controller that turns a file size into that price with the encoder in the loop, and a codec whose decoder writes a standard mesh. The result is a file of the requested size, within about 10% on four datasets, 31–122$\times$ smaller than the trained model at 0.3–1.3 dB, in one training run. The remaining gap to Gaussian codecs is the representation's; closing it, and learning the entropy model that the byte probe now measures, are the next steps.

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